Presentation Transcript — v13

Pneumonia Detection from Chest X-Rays — Classical ML vs Deep Learning, with Comparison Tests

Module: Machine Learning · Track A: Computer Vision · Unit 11

Delivery parameters

Parameter	Value
Target length	20 minutes maximum (assessment requirement)
Speaking pace	130 words per minute (sustainable conversational pace)
Slides	26 main slides (1–26) + 3 annex slides (29–31, not narrated)
Annexes	Subgroup bar charts (slide 29), training-loss curves (slide 30). Show only on question.
Demo slide	Slide 18 — embedded screen recording (~17 sec); narration explicitly references the recording

Transcript

Slide 1 — Title *(0:00 – 0:25)*

Welcome. Over the next twenty minutes I'll walk through a study detecting pneumonia in paediatric chest X-rays. The same problem tackled twice — HOG with an SVM, and ResNet50 transfer learning. Same data, same test set, same metrics. The question isn't just accuracy — it's which model to trust clinically. Let me lay out where we're going.

Slide 2 — Agenda *(0:25 – 0:45)*

Eight blocks, two worth flagging. Block one is the decision-provenance framework I used to avoid arbitrary design choices — it explains whether each key modelling decision came from literature, established practice, or direct experiment. Block seven covers two empirical tests. Let me start with the framework.

Slide 3 — Design strategy — three-tier decision-provenance framework *(1:10 – 2:05)*

Before the models, I want to make explicit how design choices were governed. This project had many degrees of freedom — image size, HOG parameters, augmentation, validation, thresholds — and not every choice has a task-specific prescription. So I used an original three-tier decision-provenance framework, informed by evidence-based practice and design-science thinking: use the strongest external evidence where it exists, use established practice or domain precedent where the evidence is indirect, and run controlled empirical comparisons where no reliable prescription exists. Tier A is direct peer-reviewed evidence. Tier B is established practice or prior task precedent — CheXNet, framework defaults. Tier C is project evidence — HOG image size, augmentation. The point is traceability: every important choice is either evidence-backed, precedent-backed, or tested.

Slide 4 — Clinical motivation *(2:05 – 2:50)*

Why this matters. Pneumonia kills over seven hundred and forty thousand children under five every year, per UNICEF. In low-resource settings the bottleneck isn't medication — it's interpretation, because radiologists are scarce. A model flagging urgent cases has real value, but the harm pathways are equally real. A missed pneumonia sends a sick child home. And — this matters for the rest of the talk — Zech and colleagues showed a chest-X-ray model latching onto chest tubes and text annotations rather than pathology. That tension is the through-line. Let me show you the data.

Slide 5 — Dataset *(2:50 – 3:30)*

The dataset is from Kermany and colleagues, twenty-eighteen — nearly six thousand paediatric chest X-rays from one hospital in Guangzhou. Roughly three pneumonia images per normal one. Two things matter. The imbalance — accuracy alone misleads, so I report macro-F1 and per-class recall. And the single-hospital limitation: one place, one age group, one country. That's not a flaw I'll hide; it becomes the spine of the ethics conversation later. With this dataset, bias and limitations aren't theoretical — they're in the metadata. Let me show you what the actual images look like.

Slide 6 — EDA — sample images *(3:30 – 4:00)*

Typical examples from each class. Notice something specific: several images carry text annotations or visible chest tubes. These aren't pathology — but they correlate with pathology, because sicker children get tubes. That correlation is the trap — a model could learn to predict the tube rather than the disease. This is exactly the concern Zech and colleagues documented, and it's why I'll pay close attention during explainability later. Now let me show you the imbalance.

Slide 7 — Class imbalance and the dataset's flawed default split *(4:00 – 4:50)*

Two issues to handle. First, the three-to-one imbalance — accuracy alone misleads, so I rely on macro-F1 and per-class recall. During training I use class-weighted loss following the inverse-frequency principle from King and Zeng. Second, the official validation folder ships with only sixteen images — far too small for reliable model selection. So I re-stratified the training pool eighty-twenty into a new train and validation, kept the official test set untouched. Stratification preserves the imbalance across folds — Kohavi nineteen ninety-five — and matters because without it a fold can land with no normal cases. Now to the preprocessing pipelines.

Slide 8 — Two preprocessing pipelines *(4:50 – 5:25)*

Two paradigms diverge on input. HOG and the SVM want grayscale, smaller images, contrast-equalised — I use CLAHE per Zuiderveld and the Pizer precursor because it limits noise amplification in dark regions. ResNet50 wants 224-pixel RGB matching ImageNet distribution, with light CheXNet-style augmentation. Both pipelines use the same train and validation splits; the test set is reserved — not used for training, validation, or model selection; used only for final evaluation and post-hoc diagnostics. That's how the comparison stays fair. Let me start with the classical model.

Slide 9 — HOG + SVM design *(5:25 – 6:15)*

Classical ML side. I deliberately picked a textbook strong pipeline, not a weak straw man — the comparison only means something if both contestants are well-built. HOG captures local edge orientations, exactly what picks up rib structure and lung consolidation — Dalal and Triggs, two-thousand-five, with the honest note that they validated on pedestrians, not medical images. RBF kernel as a standard non-linear option, balanced class weights for imbalance, and Platt scaling to give the SVM probability outputs — needed for ROC-AUC and the occlusion-sensitivity baseline. Now let's look at how I tuned this model.

Slide 10 — SVM 5-fold stratified CV — top hyperparameters *(6:15 – 7:10)*

Validation matters because if I trained one SVM, I'd have no way to know whether C and gamma were any good. Five-fold stratified cross-validation, following Kohavi nineteen ninety-five — five splits, the SVM trained five times, the average tells me which hyperparameters actually generalise. The grid is exponentially spaced for C and gamma, four values each. The search runs on a two-thousand-image subset to fit compute, then best parameters refit on the full training set. Winners on this run: C equals one, gamma equals zero point zero zero zero one, CV macro-F1 of point nine six zero nine. Now to the deep-learning side.

Slide 11 — ResNet50 — frozen-base transfer learning *(7:10 – 8:10)*

Deep-learning side. ResNet50 — He and colleagues twenty-sixteen — with ImageNet pretrained weights. That's transfer learning, the dominant paradigm in medical imaging per Cheplygina, with the principle from Yosinski: early CNN layers learn generic features that transfer well. The base is frozen — used as a feature extractor — and on top I add a small head: global average pooling per Lin and colleagues twenty-fourteen, dropout, dense, sigmoid. Head training uses Adam at learning rate one-times-ten-minus-three then refines at one-times-ten-minus-four with early stopping per Prechelt nineteen ninety-eight. One important reflection on this design choice — and I want to surface it here rather than bury it. While preparing the writeup I read the TensorFlow documentation more carefully and noticed something I hadn't understood at design time. Keras propagates the trainable flag from the parent model down to nested children, which overrides what you set on the children. My unfreeze loop was inert. What actually ran was a frozen-base feature extractor. Catching this late, with no time for a clean re-train and the recognition that the proper fix needs careful Keras nested-model work, I made the call to leave the original code as the scientific record, reframe the model honestly as a frozen-base baseline — still a standard medical-imaging approach per Cheplygina 2019 — and queue a real two-phase fine-tune as future work. The lesson sits with me: read the framework documentation before trusting a code pattern, and when you catch something late, the honest move is to document and postpone, not paper over.

Slide 12 — DenseNet121 reflection — a stronger task-specific fit (acknowledged)

(8:10 – 8:40)

This is also a Tier B lesson. I initially treated ResNet50 as the established transfer-learning baseline, but the strongest task-specific precedent — CheXNet — used DenseNet121. So ResNet50 is defensible as a canonical baseline, but not necessarily optimal for this exact task. I keep this in the talk because the framework should expose limitations, not hide them. Next iteration would compare both head-to-head from the start. Now to the training curves.

Slide 13 — CNN training curves and 5-fold cross-validation *(8:55 – 9:50)*

Training curves on the left, five-fold CV on the right. The head training schedule — learning rate one-times-ten-minus-three then one-times-ten-minus-four — early-stopped, train and validation stay close throughout. Five-fold CV anchors in Kohavi nineteen ninety-five — same Tier A backing as the SVM. Mean macro-F1 is zero point eight nine three three, standard deviation zero point zero three zero nine. The fold range was zero point eight three to zero point nine two three — real variance worth reporting. This is a compute-light five-fold validation of the CNN pipeline — frozen base, four epochs per fold — and the headline model is the same frozen-base ResNet50 with the head learning-rate schedule, so the CV is a pipeline robustness check. The CNN's 5-fold CV mean is zero point eight nine three three with standard deviation zero point zero three zero nine. The held-out test result is zero point seven seven two one — a twelve-point drop. The SVM's CV is zero point nine six zero nine dropping to zero point six eight five on test, a twenty-seven-point drop. So the CNN's gap is no longer near-zero, but it is roughly two times cleaner than the SVM's. That comparative point still grounds the recommendation. Now to the head-to-head.

Slide 14 — Test-set head-to-head — held-out 624 images *(9:50 – 10:55)*

This is the heart of what I set out to do — head to head, which paradigm wins? Macro-F1: zero point seven seven two one for the CNN against zero point six eight four six for the SVM, about a nine-point gap on the metric imbalance can't fool. NORMAL recall: zero point five one for the CNN, zero point three seven for the SVM. The most striking signal is the validation-versus-test comparison. The SVM's validation score was zero point nine six zero nine but its test score is zero point six eight four six — a twenty-seven-point generalisation gap. The CNN's pipeline CV was zero point eight nine three three and its test zero point seven seven two one — a twelve-point gap, which is meaningful but far tighter than the SVM's spread. The deep-learning pipeline generalises more robustly than HOG. One honest caveat: I haven't tested whether the nine-point CNN-versus-SVM gap is statistically significant; bootstrap CI or McNemar's would be future work. Now to where the errors fall.

Slide 15 — Confusion matrices — and a note on the threshold *(10:55 – 11:40)*

Drilling into where each model errs. The bottom-left cell is what matters clinically — pneumonia we missed. Both close to perfect there: SVM caught three hundred and eighty-six of three hundred and ninety pneumonias, CNN caught three hundred and eighty-eight. But look at the normals row. The SVM identified only eighty-seven of two hundred and thirty-four normals correctly — misclassifying one hundred and forty-seven healthy children as pneumonia. The CNN correctly classified a hundred and twenty of two hundred and thirty-four NORMAL cases, matching the v12 NORMAL recall of zero point five one. So the SVM's pneumonia recall comes from heavily over-predicting pneumonia, not from genuine class separation — and these are the numbers at HOG_SIZE=128. The threshold of zero point five is the reporting default, which I revisit concretely with measured numbers in ethics. Now to what the models are actually looking at.

Slide 16 — Explainability — HOG-viz + occlusion sensitivity (SVM) *(11:40 – 12:20)*

For a clinical model we need to know not just whether it's right, but why. I take a deliberately simpler unified XAI approach: HOG visualisation for the feature representation, plus occlusion sensitivity for the decision drivers. HOG visualisation comes from Dalal and Triggs twenty-oh-five, built into scikit-image — it shows the gradient orientations the SVM is operating on, directly. Occlusion sensitivity is Zeiler and Fergus twenty-fourteen — slide a small grey patch across the image, record how the predicted pneumonia probability drops at each position; where it drops most is where the SVM relied. On the correctly classified PNEUMONIA — bacteria_475 — the SVM is at one-point-zero saturation, so the occlusion heatmap range is small. That's a known limitation of occlusion sensitivity when the baseline is at the boundary. On the misclassified NORMAL — IM-0001-0001, confidently called PNEUMONIA at point nine seven — the heatmap shows the strongest signal on shoulder and right-edge regions outside the lungs, consistent with the Zech twenty-eighteen shortcut-learning risk but with small dynamic range limiting what we can conclude from saturation. I picked HOG-viz plus occlusion as the simpler unified pair that works identically on the CNN side. Now to the CNN side.

Slide 17 — Explainability — occlusion sensitivity (CNN) *(12:20 – 13:00)*

Slide seventeen — explainability for the CNN, occlusion sensitivity. Both example cases on this CNN — the correctly classified PNEU and the misclassified NORMAL — saturated at probability one point zero zero zero zero in the headline trained instance. That means the heatmap dynamic range collapses to roughly two-thousandths of a probability point on each panel. The pattern that does emerge is that on both cases the CNN attends most strongly to the shoulder and edge regions, not to the lung field. That is consistent with the shortcut-learning concern Zech and colleagues raised in two thousand eighteen, but the saturated baseline (both at P=1.0000, dynamic range ~0.002) means we should treat this as a warning signal motivating richer XAI, not conclusive evidence. It also reinforces the slide-twenty-five calibration concern, because the CNN's sigmoid is producing one point zero zero zero zero on a NORMAL image it gets confidently wrong.

Slide 18 — Model demonstration — one held-out test X-ray *(13:00 – 13:55)*

Slide eighteen — demonstration on a held-out NORMAL X-ray. The video shows both models running end-to-end on NORMAL-two-I-M-zero-one-three-five-zero-zero-zero-one, a paediatric NORMAL chest X-ray that was held out from training and hyperparameter tuning. Both models load from the persisted weights — the same trained instance that produced the headline metrics on slide fourteen and the synthesis on slide twenty-four. The SVM's HOG-CLAHE-scaler-Platt pipeline produces a probability of zero point nine-five-four-nine, which crosses the default zero-point-five threshold and predicts PNEUMONIA. The CNN's ResNet50 frozen-base pipeline produces a probability of zero point three-seven-eight-nine, which falls below the threshold and predicts NORMAL. The true label, revealed after the prediction, is NORMAL. So on this single image, the SVM is confidently wrong and the CNN is correctly classifying with moderate confidence. This is the empirical illustration of the SVM-as-shadow-model design on slide twenty-five: when the cheap SVM says PNEUMONIA and the CNN says NORMAL on the same image, the disagreement itself is the signal a radiologist should investigate.

Slide 19 — Subgroup performance — three intrinsic splits *(13:50 – 14:40)*

Aggregate accuracy can hide disparities. Three intrinsic splits. Viral subtype: both models catch nearly every pneumonia case — CNN PNEUMONIA recall one point zero, SVM zero point nine nine three. Bacterial subtype: both models tied, both above zero point ninety-nine. These subgroups are single-class pneumonia, so macro-F1 collapses to about zero point five — that is a metric artefact, not a real finding. Read PNEUMONIA recall instead: CNN and SVM are near-identical in clinical performance. Brightness terciles — the SVM drops fourteen points from zero point seven four at low brightness to zero point six at high brightness; the CNN nine points from zero point eight zero to zero point seven one. The CNN is more robust to brightness variation, but both models lose ground at high brightness. Image-size terciles: both models are tied at zero point five on small images — small images are hardest for both. The CNN outperforms the SVM on mid-size images, zero point six three versus zero point five six, and on large images, zero point five eight versus zero point four eight. External-validity caveat: Zech and colleagues saw a chest-X-ray model drop significantly on a different hospital's data; that hospital-shift gap is hidden from a single-site study like this one. Now to the experiments I ran when no external answer existed.

Slide 20 — Comparison test 1 — HOG image size *(14:40 – 15:25)*

First of two design tests. Question: what HOG image size? No clear external answer, so I tested sixty-four, one twenty-eight, and one ninety-two pixels, holding the SVM hyperparameters constant. Selection was done on validation data only, then reported on the test set to avoid leakage. The validation macro-F1: zero point nine five eight at sixty-four, zero point nine six four at one twenty-eight, zero point nine six one at one ninety-two. One twenty-eight wins on validation, by a small but consistent margin. The headline SVM pipeline uses one twenty-eight, anchored in a clean comparison test. One caveat: the comparison holds SVM hyperparameters fixed across sizes; different feature dimensions probably want different C and gamma, so a fully principled comparison would re-tune per size. That's documented as a limitation and listed in next steps. This is Tier C doing its job: where I had no reliable external prescription, I generated project evidence cleanly and used the result in the headline pipeline.

Slide 21 — Comparison test 2 — CNN augmentation on/off *(15:25 – 15:55)*

Second design test, CNN side, and this is another Tier C result — but a different kind. I ran the test twice and got opposite orderings, both within typical run-to-run variance for single-seed ResNet50 head training. Bouthillier and colleagues document exactly this. The empirical test did not give a stable winner; it told me the evidence was underpowered at one seed per arm. So the framework does not just justify decisions — it also tells me when the evidence is not strong enough to settle the question; multi-seed averaging is what would strengthen it. Reporting both runs beats cherry-picking. Now to ethics.

Slide 22 — Ethics — grounded in this dataset *(16:20 – 17:15)*

Ethics, grounded in this dataset's specific problems. Three columns. Privacy: paediatric, single hospital, retrospective. DICOM Supplement 142 strips header identifiers, but text burned into image pixels bypasses that — any deployment needs OCR-based pixel-text redaction. Public Kaggle redistribution doesn't re-state consent — fresh governance review unavoidable. Fairness and error costs: a missed pneumonia is potentially fatal, a false flag costs minutes. From the actual threshold sweep on this CNN: at the default zero point five we get one hundred and fourteen false positives and two missed pneumonias, with PNEU recall at zero point nine nine four nine. RAISING the threshold to zero point eighty reduces false positives to eighty-four — that is thirty fewer false positives — at the cost of one extra missed pneumonia, FN: two to three. PNEU recall stays at zero point nine nine two three, still above zero point nine nine. The trade-off direction is fewer false positives per extra missed pneumonia

— a measured cost-benefit a clinician or governance committee would weigh, not one we accept on their behalf. That trade-off should be a clinician decision, not a default. Bias and harm: paediatric to adult silently degrades; equipment shift changes inputs; brightness sensitivity is real on both models; shortcut learning is something the occlusion heatmap helps check visually. Each is a deployment veto without active monitoring. Speaking of deployment.

Slide 23 — Deployment architecture — UML deployment / component diagram

(17:15 – 18:00)

Slide twenty-three — deployment architecture. The diagram has three named planes. First, the hospital data plane. PACS as the DICOM source feeds an image adapter that converts to JPEG and rejects adult X-rays as out-of-distribution. The adapter output flows into the cloud, and the result returns to the clinician worklist UI showing the predicted probability and an on-demand region-importance heatmap. Second, the cloud inference plane. The triage endpoint runs the ResNet50 frozen-base model as the primary predictor with the SVM as a transparent shadow comparator. The model registry stores versioned dot-keras artefacts with config and content hash — that is Decision twenty-one operationalised. The audit log captures every prediction with the input hash, the model version, and the clinician's verdict. Third, the MLOps operations plane. The drift monitor runs a daily Kolmogorov-Smirnov test on luminance and image-size distributions. The performance monitor evaluates weekly macro-F1 on a relabelled sample plus subgroup metrics across brightness, subtype, and equipment vendor. The retraining trigger fires on a KS breach lasting seven days, a macro-F1 drop greater than three percentage points, or a new equipment vendor entering the stream. The retraining pipeline produces a new artefact but requires human review before promotion to the registry — that is the governance gate. The agent terminology and the boundary structure follow Sculley and colleagues, twenty-fifteen, on hidden technical debt in machine learning systems. The governance plane on the right summarises the operational constraints: clinician-selected operating point from the slide-twenty-two threshold sweep, out-of-distribution rejection for adult X-rays, and the Class Two-A regulatory frame under MDR Annex Eight Rule Eleven.

Slide 24 — Classical ML vs Deep Learning, comparative synthesis *(17:50 – 18:10)*

This slide pulls the comparison together at a higher level. Conceptually, the SVM relies on hand-crafted HOG features over an eight-thousand-one-hundred-dimensional descriptor; the CNN learns hierarchical features directly from two-hundred-twenty-four-pixel RGB inputs through transfer learning. In practice that translated into a nine-point macro-F1 win for the CNN, a fourteen-point NORMAL-recall win, and a generalisation gap roughly two times cleaner than the SVM's. Both models surfaced shoulder-and-edge attention on hard cases under occlusion sensitivity, which is consistent with the Zech 2018 shortcut-learning risk but the saturated baseline (both at P=1.0000, dynamic range ~0.002) means we should treat this as a warning signal rather than conclusive evidence. Operationally, the SVM remains useful as a transparent baseline and disagreement-flagging shadow model; the CNN is the recommended primary candidate, under the safeguards on the next slide.

Slide 25 — Recommendation — ResNet50 with safeguards *(17:50 – 18:50)*

Pulling it together — deploy ResNet50, with safeguards. Why ResNet over the SVM: a nine-point macro-F1 gap, NORMAL recall of zero point five one against zero point three seven, and generalization robustness where the CNN has a twelve-point CV-test gap and the SVM has a twenty-seven-point gap — the CNN generalises roughly two times more cleanly than the SVM, not perfectly, but the comparative point holds. Why I'd keep the SVM: cheap to run as a shadow model, disagreements flag cases for radiologist review, fully inspectable for regulators. Five non-negotiable safeguards before anything near a patient. One, the operating threshold is a clinician decision, not a hard-coded constant — with a documented target such as pneumonia recall at zero point nine nine if the false-positive workload is acceptable, as the threshold sweep on slide twenty-two showed. Two, region-importance heatmap available on demand for clinician review of borderline cases; lighter Grad-CAM is the planned routine UI surface — occlusion is roughly one hundred times more expensive per explanation than Grad-CAM at inference time, so for live triage volume the cost difference matters. Three, weekly subgroup monitoring across brightness, disease subtype, equipment vendor. Four, explicit rejection of adult X-rays as out-of-distribution. Five, human-in-the-loop always. Without all five, this stays a prototype. Now into reflection.

Slide 26 — Critical reflections — what I would do differently *(18:50 – 19:35)*

Stepping back, the framework made the project more auditable, but it also exposed where the evidence was weaker — and where I deliberately chose simpler over richer. The most informative finding is the generalisation-gap signal: SVM dropping twenty-seven points from validation to test while the CNN has a twelve-point CV-test gap and the SVM has a twenty-seven-point gap — the CNN generalises roughly two times more cleanly. That's the headline insight I'd lead with next time. The frozen-base design reframing taught me a lesson about reading documentation carefully. While preparing the writeup I read the TensorFlow documentation more carefully and noticed something I hadn't understood at design time. Keras propagates the trainable flag from the parent model down to nested children, which overrides what you set on the children. My unfreeze loop was inert. What actually ran was a frozen-base feature extractor. Catching this late, with no time for a clean re-train and the recognition that the proper fix needs careful Keras nested-model work, I made the call to leave the original code as the scientific record, reframe the model honestly as a frozen-base baseline — still a standard medical-imaging approach per Cheplygina 2019 — and queue a real two-phase fine-tune as future work. The lesson sits with me: read the framework documentation before trusting a code pattern, and when you catch something late, the honest move is to document and postpone, not paper over. Looking back, I learned that trained-model artefacts are first-class versioned outputs, not transient session state. The original v8 weights were not preserved as a file, so when reviewer feedback prompted swapping the XAI tool, the only path forward was a re-train, which split the evidence chain. The fix was a single end-to-end re-run; the prevention going forward is disciplined model-weight persistence. Saving the model is methodology, not housekeeping. I came into this project without prior experience in image processing or chest-X-ray imaging. To make any of the design choices defensible — the image resize size, the CLAHE contrast preprocessing, the HOG cell parameters, the augmentation policy, the subgroup splits, the entire ethics framing around shortcut learning — I had to invest learning time in two domain layers first. Image-processing fundamentals: pixels, resolution, what HOG features are actually capturing. Chest-X-ray specifics: what bacterial and viral pneumonia look like radiographically, what non-pathological artefacts create shortcut-learning risk, which subpopulations the dataset systematically under-represents. Without that learning the code would still run, but every design choice would be arbitrary. The Tier C category in my decision-provenance framework exists exactly for this — empirically grounded choices made by someone who has learned to see what is in the dataset. The simpler-XAI-by-design choice: I deliberately picked unified occlusion sensitivity plus HOG visualisation over richer attribution methods. Same technique on both models, no callable shim complexity — and crucially, a unified comparative narrative that surfaces the same shortcut-learning pattern across paradigms. Richer methods remain as planned refinements. Finally, a single-seed augmentation comparison wasn't enough — multi-seed from the start would have been more rigorous. Honest reflection beats post-hoc justification. Closing reflection. What felt natural from my enterprise-architecture background was the systems side: traceability, governance, deployment risk, monitoring, and model-artefact control. The decision-provenance framework I introduced on slide 3 is that architecture instinct applied to ML uncertainty — a way of making every modelling choice auditable rather than treating them as opaque expert judgement. What I genuinely had to learn was ML-specific: validation leakage, stochastic CNN training behaviour, frozen-base transfer learning, the limits of XAI when probabilities saturate, calibration, and disciplined model-weight persistence. That is why the final recommendation is cautious. ResNet50 here is the stronger triage candidate — not a clinical product. External validation on a different cohort, weekly monitoring, clinician-selected threshold governance, and human-in-the-loop oversight are non-negotiable before any patient-facing use.

Slide 27 — Limitations and what I would do next *(19:35 – 20:00)*

Eight concrete next steps. External validation on a different cohort — RSNA or NIH ChestX-ray14. DenseNet121 head-to-head against ResNet50. A true two-phase fine-tune fixing the Keras nested-model trainable-flag issue. Per-size HOG hyperparameter retuning — the size comparison held C and gamma fixed across sizes; a fully principled comparison would re-tune per size. Add LIME on the SVM and Grad-CAM on the CNN as richer attribution methods beyond the current occlusion sensitivity baseline. Multi-seed averaging on the comparison tests. Calibration analysis: reliability curves, Expected Calibration Error (ECE), Brier score, and temperature scaling for the CNN; calibration check on Platt-scaled SVM outputs; bootstrap CI on macro-F1 and McNemar's test on paired predictions. And prospective rather than retrospective evaluation — every number here is retrospective; a clinical claim requires prospective data. Thank you. Happy to take questions.

Where to slow down for emphasis

- Slide 12 (DenseNet121 reflection) — visible critical thinking marker for the examiner.
- Slide 14 (CNN versus SVM finding) — the headline comparative insight in the deck.
- Slide 17 (occlusion sensitivity on the CNN) — unified XAI; same technique as slide 16 SVM.
- Slide 20 (comparison test 2 — augmentation) — empirical demonstration of augmentation as regulariser.
- Slide 24 (critical reflections) — second moment of demonstrated metacognition.

Where to speed up if behind

- Slide 8 (preprocessing) — compresses cleanly without losing substance.
- Slide 22 (deployment) — already concise.
- Slide 27 (limitations) — can be trimmed in delivery if needed.

Verbal cues to add live

- "Look at the green row" on slides 19–20 (comparison test results).
- "See the bottom-left cell" on slide 15 (confusion matrices).
- "Same XAI tool, both paradigms" on slides 16–17 (occlusion sensitivity unified).
- "See the red bar on the left" on slide 25 (reflections cards).

What does NOT get narrated

- Slide 28 (References) — leave on screen briefly so the marker can verify citations match.
- Slide 29 (Annex divider) — silent.
- Slides 30–31 (Subgroup detail, training loss) — only show if a question demands them.

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